DevOps



LAB 03

Working with Git History, Stashing, and Reverting Commits

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BCS 7-B

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Task 1: Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)

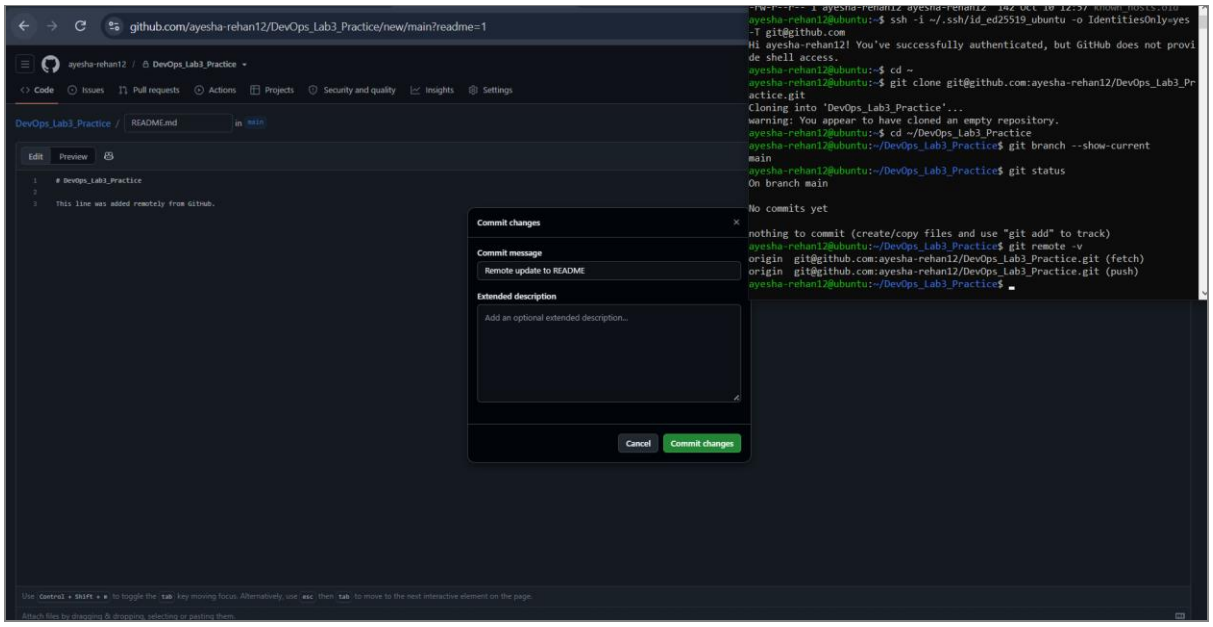


Figure: README.md edited and committed on GitHub (remote change).

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git remote -v
origin  git@github.com:ayesha-rehan12/DevOps_Lab3_Practice.git (fetch)
origin  git@github.com:ayesha-rehan12/DevOps_Lab3_Practice.git (push)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local update to README"
[main (root-commit) 2860943] Local update to README
 1 file changed, 4 insertions(+)
 create mode 100644 README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: README.md edited locally and committed.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:ayesha-rehan12/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Push rejected because the remote contains work not present locally.

```
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git pull --no-rebase origin main
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 922 bytes | 1024 bytes/s, done.
From github.com:ayesha-rehan12/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
fatal: refusing to merge unrelated histories
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: git pull --no-rebase merging remote and local commits into a merge commit.

```
nothing to commit, working tree clean
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice

This line was added locally.
This line was added remotely from GitHub.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 577 bytes | 57.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
   f043c29..5abf20e  main -> main
branch 'main' set up to track 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Successful push after the merge.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Another local update to README"
[main d59e330] Another local update to README
 1 file changed, 2 insertions(+)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:ayesha-rehan12/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 998 bytes | 249.00 KiB/s, done.
From github.com:ayesha-rehan12/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   7822ba3..bd9fa3f  main       -> origin/main
Auto-merging README.md
```

Figure: git pull --rebase replaying local commits on top of remote commits.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 339 bytes | 339.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
   bd9fa3f..d9154e7  main -> main
branch 'main' set up to track 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Successful push after the rebase, with linear history.

Task 2: Creating and Resolving Merge Conflicts Manually

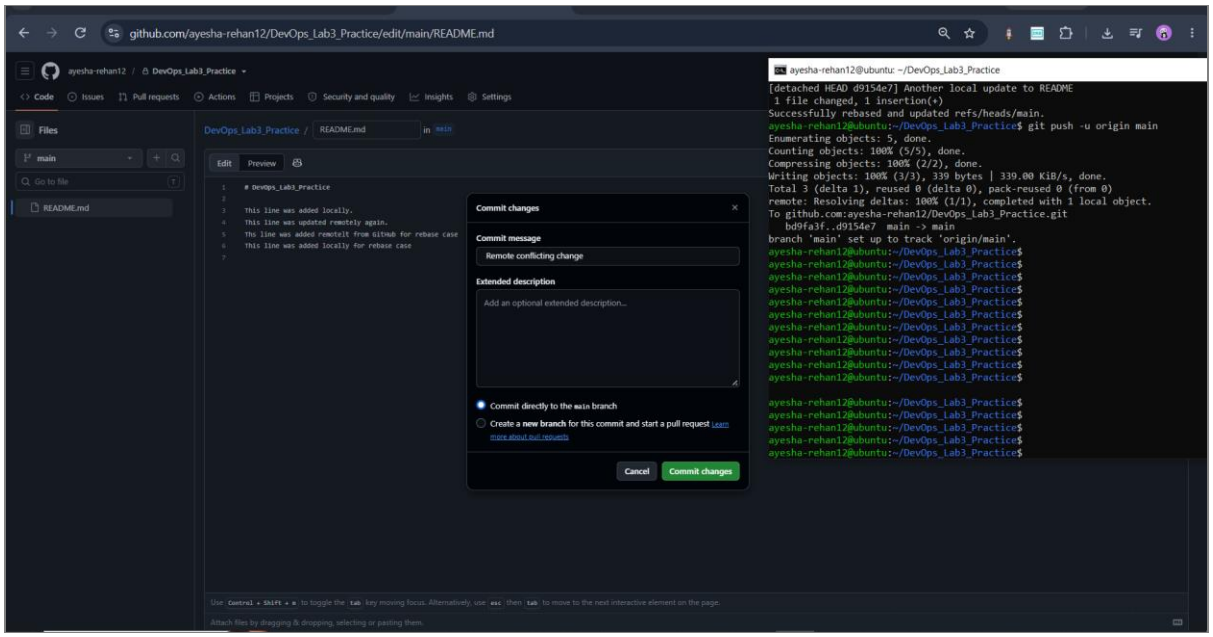


Figure: Conflicting change made to the same line on GitHub.



Figure: Same line edited differently locally in VS Code.

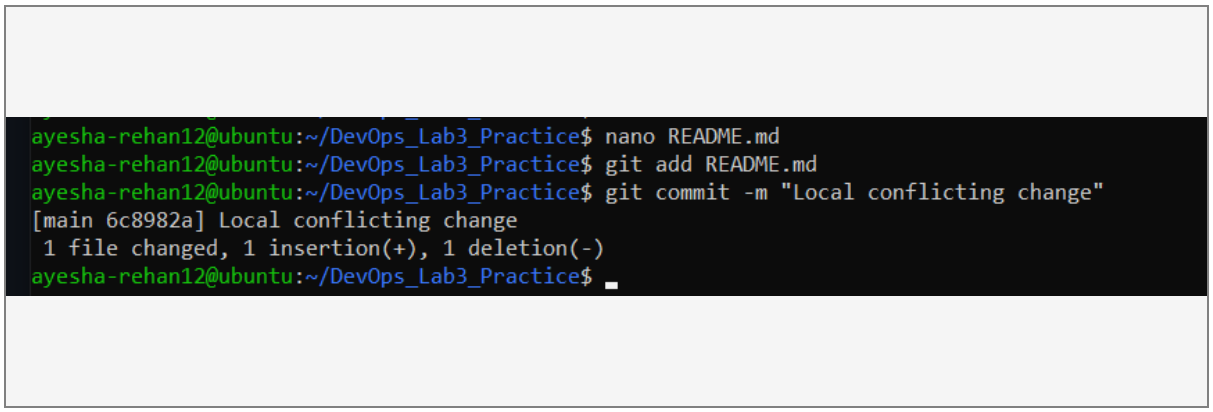


Figure: Local conflicting change staged and committed.

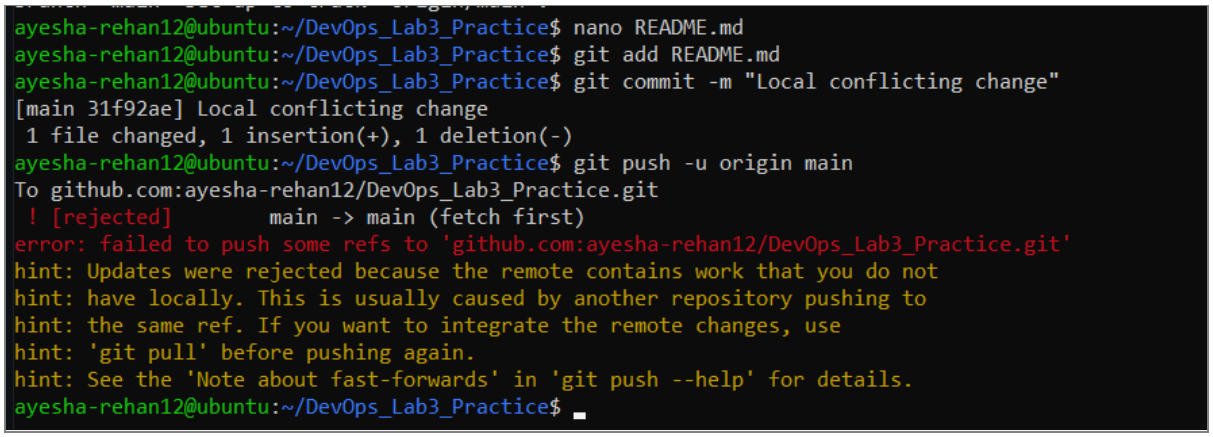


Figure: Push rejected due to conflicting remote changes.

```
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 937 bytes | 468.00 KiB/s, done.
From github.com:ayesha-rehan12/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   6c8982a..0400e15  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply 31f92ae... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 31f92ae... # Local conflicting change
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Rebase paused with a conflict message after git pull --rebase.

```
ayesha-rehan12@ubuntu: ~/DevOps_Lab3_Practice
  GNU nano 8.7.1                                README.md
# DevOps_Lab3_Practice

This line was added locally.
This line was updated remotely.
This line was updated locally at the same time.
Ths line was added remotelt from GitHub for rebase case
This line was added locally for rebase case

                          [ Wrote 7 lines ]
^G Help       ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

Figure: README.md with conflict markers manually resolved.

```
Could not apply 31f92ae... # Local conflicting change
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git rebase --continue
[detached HEAD 78ed8d9] Local conflicting change
 1 file changed, 2 insertions(+), 1 deletion(-)
Successfully rebased and updated refs/heads/main.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Conflict marked resolved with git add and git rebase --continue.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git rebase --continue
[detached HEAD 78ed8d9] Local conflicting change
 1 file changed, 2 insertions(+), 1 deletion(-)
Successfully rebased and updated refs/heads/main.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 321 bytes | 321.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
   0400e15..78ed8d9  main -> main
branch 'main' set up to track 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Resolved changes pushed to remote.

Task 3: Managing Ignored Files with .gitignore and Removing Tracked Files

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ mkdir textfiles
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ echo "File A content" > textfiles/a.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ echo "File B content" > textfiles/b.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ echo "File C content" > textfiles/c.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add .
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added textfiles directory with three files"
[main 3270bab] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 463 bytes | 115.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
   78ed8d9..3270bab  main -> main
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: textfiles directory with three files committed and pushed.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ echo "textfiles/*" > .gitignore
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add .gitignore
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added .gitignore to ignore textfiles directory"
[main 9e2e640] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 350 bytes | 350.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
   3270bab..9e2e640  main -> main
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: .gitignore added and pushed.

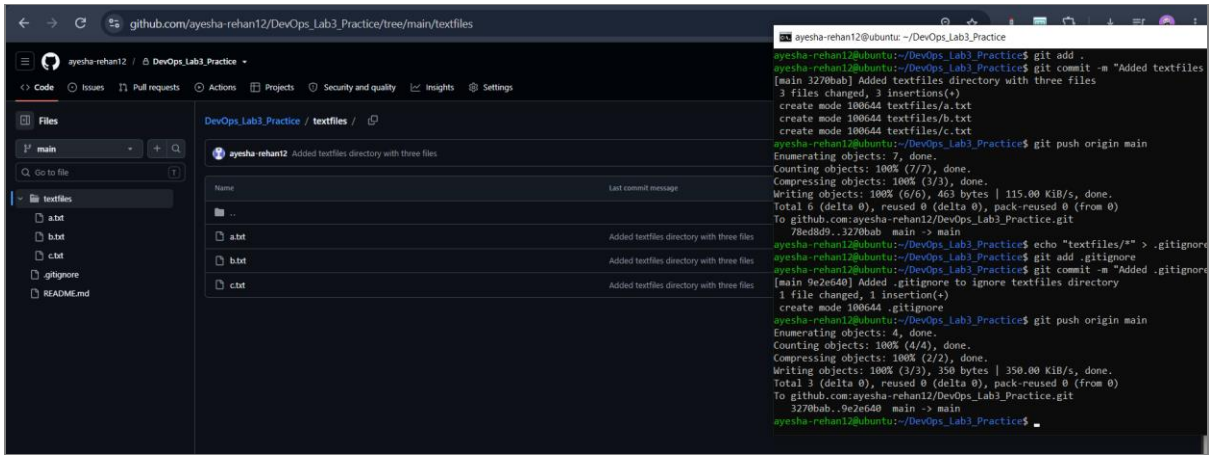


Figure: GitHub repository still showing the textfiles directory.

```
   3270bab..9e2e640  main -> main
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add .
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Removed tracked textfiles directory"
[main e7e33ae] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 240 bytes | 240.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
   9e2e640..e7e33ae  main -> main
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Files untracked with git rm -r --cached and pushed.

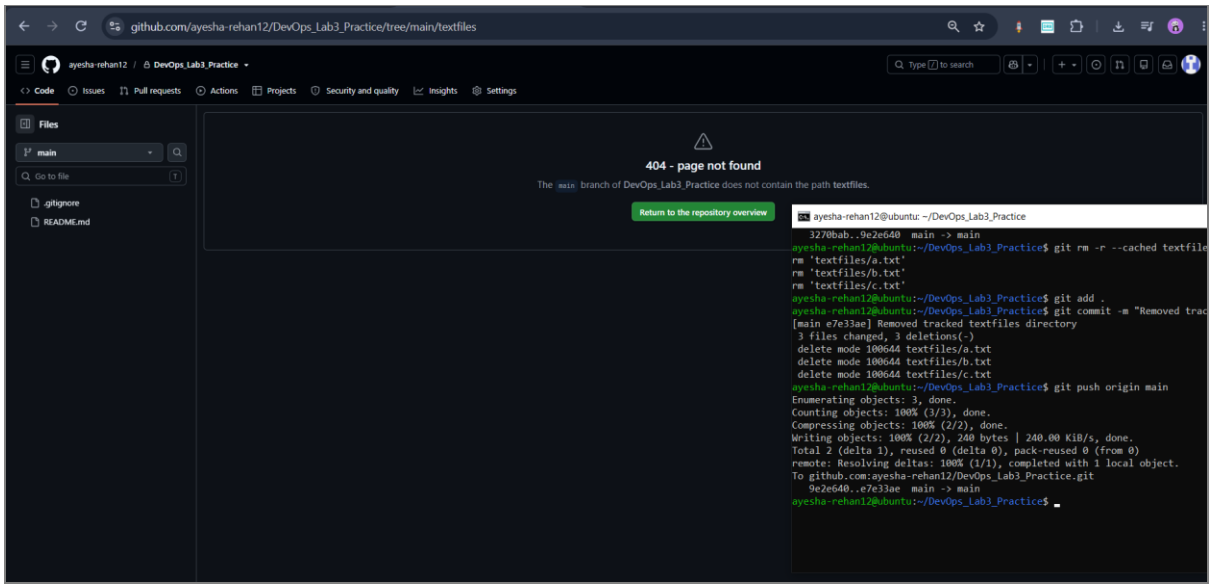


Figure: GitHub repository after textfiles folder was removed remotely.

Task 4: Create Temporary Changes and Use git stash

```
nothing to commit, working tree clean
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git branch
* main
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Changes the README.md file"
[feature-branch a936688] Changes the README.md file
 1 file changed, 2 insertions(+)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: README.md modified and committed on feature-branch.

```
 1 file changed, 2 insertions(+)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Checkout to main blocked due to uncommitted local changes.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git stash
Saved working directory and index state WIP on feature-branch: a936688 Changes the README.md file
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Uncommitted changes saved with git stash.

```
Aborting
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git stash
Saved working directory and index state WIP on feature-branch: a936688 Changes the README.md file
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Successfully switched to main after stashing.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Switched back to feature-branch.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch feature-branch
nothing to commit, working tree clean
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: git status showing working tree clean.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (660caf2a263293f1b2943fd410836eb7062f7b25)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Stashed changes restored with git stash pop.

Task 5: Checkout a Specific Commit Using git log

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
a936688 (HEAD -> feature-branch) Changes the README.md file
e7e33ae (origin/main, main) Removed tracked textfiles directory
9e2e640 Added .gitignore to ignore textfiles directory
3270bab Added textfiles directory with three files
78ed8d9 Local conflicting change
0400e15 Remote conflicting change
6c8982a Local conflicting change
d9154e7 Another local update to README
bd9fa3f Update README.md for rebase scenario
7822ba3 Local update to README for rebase scenario
03d86e6 Update README.md remotely for rebase scenario
5abf20e Merge remote and local README histories
f043c29 Remote update to README
2860943 Local update to README
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Commit history shown with git log --oneline.

```
ayesha-rehan12@ubuntu: ~/DevOps_Lab3_Practice
3270bab Added textfiles directory with three files
78ed8d9 Local conflicting change
0400e15 Remote conflicting change
6c8982a Local conflicting change
d9154e7 Another local update to README
bd9fa3f Update README.md for rebase scenario
7822ba3 Local update to README for rebase scenario
03d86e6 Update README.md remotely for rebase scenario
5abf20e Merge remote and local README histories
f043c29 Remote update to README
2860943 Local update to README
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout 78ed8d9
Note: switching to '78ed8d9'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 78ed8d9 Local conflicting change
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Checked out a specific commit, showing detached HEAD state.

```
HEAD is now at 78ed8d9 Local conflicting change
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Previous HEAD position was 78ed8d9 Local conflicting change
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Returned to the main branch.

Task 6: Resetting Commits (Soft vs Hard Reset)

```
nothing to commit, working tree clean
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add .
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added test line"
[main e09e947] Added test line
 1 file changed, 1 insertion(+)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: First test line added and committed.

```
 1 file changed, 1 insertion(+)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add .
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second test commit"
[main 280c313] Second test commit
 1 file changed, 1 insertion(+)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Second test commit made.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
280c313 (HEAD -> main) Second test commit
e09e947 Added test line
e7e33ae (origin/main) Removed tracked textfiles directory
9e2e640 Added .gitignore to ignore textfiles directory
3270bab Added textfiles directory with three files
78ed8d9 Local conflicting change
0400e15 Remote conflicting change
6c8982a Local conflicting change
d9154e7 Another local update to README
bd9fa3f Update README.md for rebase scenario
7822ba3 Local update to README for rebase scenario
03d86e6 Update README.md remotely for rebase scenario
5abf20e Merge remote and local README histories
f043c29 Remote update to README
2860943 Local update to README
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Commit history before reset.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice

This line was added locally.
This line was updated remotely.
This line was updated locally at the same time.
Ths line was added remotelt from GitHub for rebase case
This line was added locally for rebase case
Task 6 first test line.
Task 6 second test line.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: File containing both added lines before reset.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git reset --soft HEAD~1
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: git reset --soft HEAD~1 performed.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
e09e947 (HEAD -> main) Added test line
e7e33ae (origin/main) Removed tracked textfiles directory
9e2e640 Added .gitignore to ignore textfiles directory
3270bab Added textfiles directory with three files
78ed8d9 Local conflicting change
0400e15 Remote conflicting change
6c8982a Local conflicting change
d9154e7 Another local update to README
bd9fa3f Update README.md for rebase scenario
7822ba3 Local update to README for rebase scenario
03d86e6 Update README.md remotely for rebase scenario
5abf20e Merge remote and local README histories
f043c29 Remote update to README
2860943 Local update to README
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Commit history after soft reset.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice

This line was added locally.
This line was updated remotely.
This line was updated locally at the same time.
Ths line was added remotelt from GitHub for rebase case
This line was added locally for rebase case
Task 6 first test line.
Task 6 second test line.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: File still containing both edits after soft reset.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at e09e947 Added test line
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: git reset --hard HEAD~1 performed.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
e09e947 (HEAD -> main) Added test line
e7e33ae (origin/main) Removed tracked textfiles directory
9e2e640 Added .gitignore to ignore textfiles directory
3270bab Added textfiles directory with three files
78ed8d9 Local conflicting change
0400e15 Remote conflicting change
6c8982a Local conflicting change
d9154e7 Another local update to README
bd9fa3f Update README.md for rebase scenario
7822ba3 Local update to README for rebase scenario
03d86e6 Update README.md remotely for rebase scenario
5abf20e Merge remote and local README histories
f043c29 Remote update to README
2860943 Local update to README
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Commit history after hard reset.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice

This line was added locally.
This line was updated remotely.
This line was updated locally at the same time.
Ths line was added remotelt from GitHub for rebase case
This line was added locally for rebase case
Task 6 first test line.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: File after hard reset with the second edit removed.

Task 7: Amending the Last Commit

```
nothing to commit, working tree clean
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add .
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Fix log message"
[main 2c22107] Fix log message
 1 file changed, 1 insertion(+)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Initial commit before amending.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add .
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit --amend
[main ca64ed3] Fix log message
 Date: Sun Oct 18 04:44:03 2026 +0000
 1 file changed, 2 insertions(+)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Last commit amended with git commit --amend.

Task 8: Reverting a Commit (Safe Undo on Remote Branch)

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ echo "temporary text" >> temp.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add .
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added temporary text file"
[main ec6edd7] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 914 bytes | 228.00 KiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
   e7e33ae..ec6edd7  main -> main
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: temp.txt committed and pushed to main.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git revert ec6edd7
[main fc5e848] Revert "Added temporary text file"
 1 file changed, 1 deletion(-)
 delete mode 100644 temp.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Commit reverted with git revert <commit-hash>.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 257 bytes | 257.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
   ec6edd7..fc5e848  main -> main
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Revert commit pushed to remote.

Task 9: Force Push (With Caution)

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b test-force
Switched to a new branch 'test-force'
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: New branch test-force created.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ echo "Task 9 force-push demonstration." >> force-demo.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git add force-demo.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Add force push demo change"
[test-force 818bbd2] Add force push demo change
 1 file changed, 1 insertion(+)
 create mode 100644 force-demo.txt
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Small change committed on test-force.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 363 bytes | 121.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/ayesha-rehan12/DevOps_Lab3_Practice/pull/new/test-force
remote:
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
 * [new branch]      test-force -> test-force
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: test-force branch pushed to remote.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at fc5e848 Revert "Added temporary text file"
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Hard reset performed with git reset --hard HEAD~1.

```
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 363 bytes | 121.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/ayesha-rehan12/DevOps_Lab3_Practice/pull/new/test-force
remote:
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
 * [new branch]      test-force -> test-force
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at fc5e848 Revert "Added temporary text file"
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:ayesha-rehan12/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Normal push rejected by the remote repository.

```
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ayesha-rehan12/DevOps_Lab3_Practice.git
 + 818bbd2...fc5e848 test-force -> test-force (forced update)
ayesha-rehan12@ubuntu:~/DevOps_Lab3_Practice$
```

Figure: Force push to test-force succeeded.

Task 10: Running Gitea in GitHub Codespaces via Docker Compose

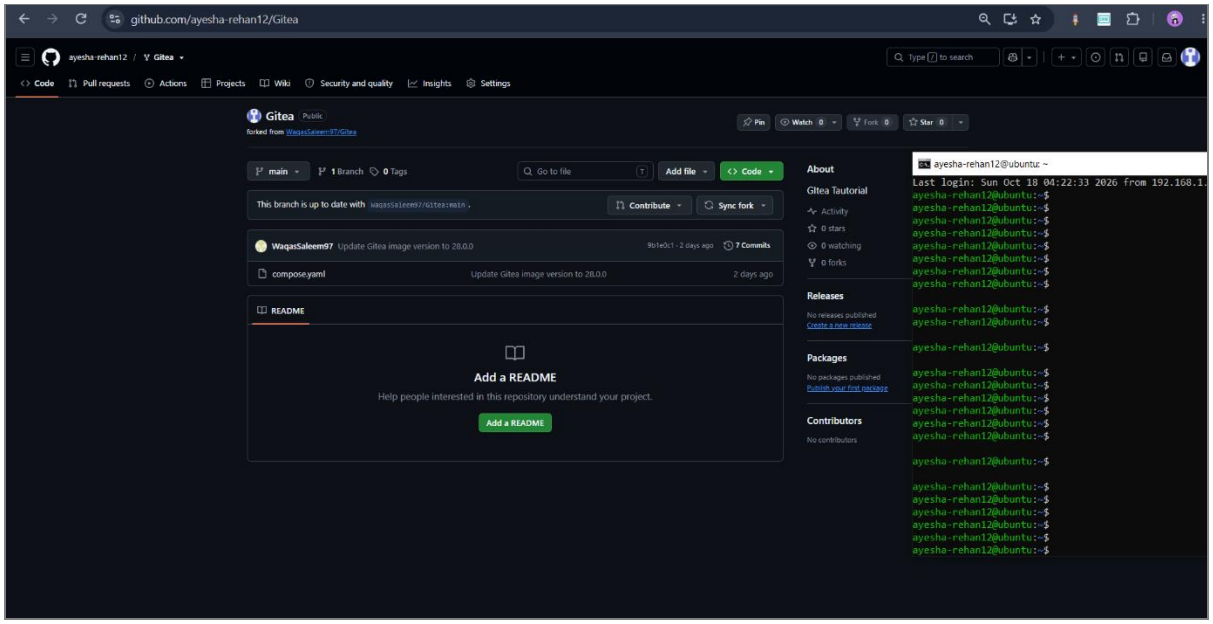


Figure: Gitea repository forked into my GitHub account.

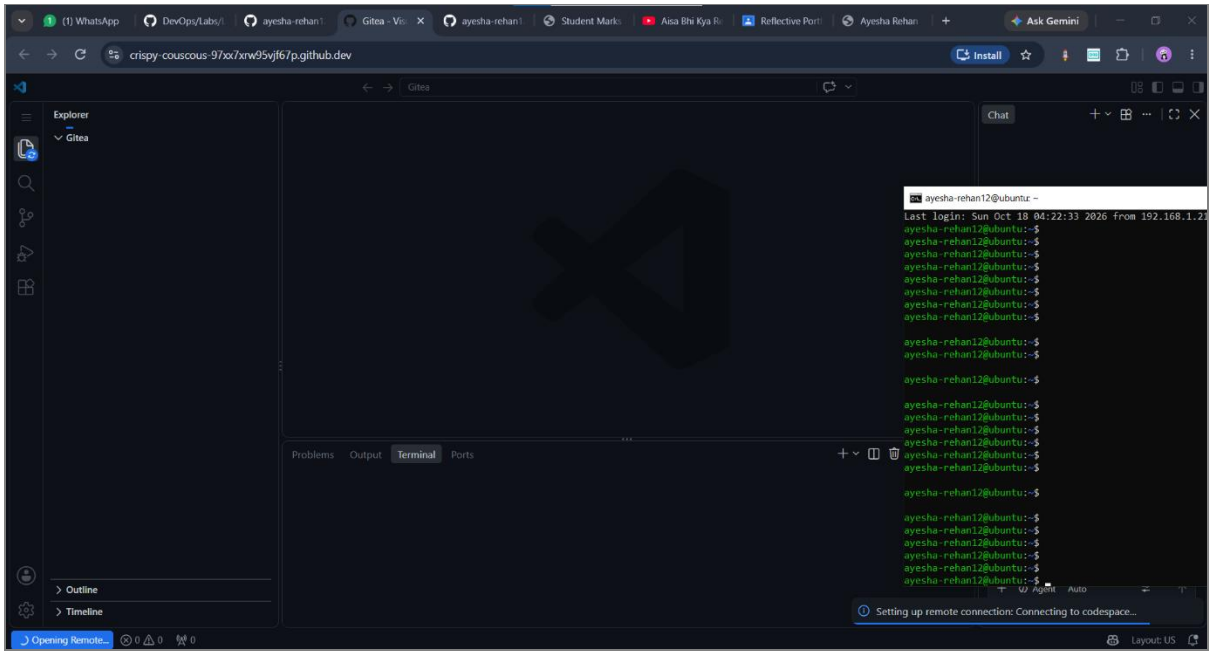


Figure: GitHub Codespace loading for the forked repository.

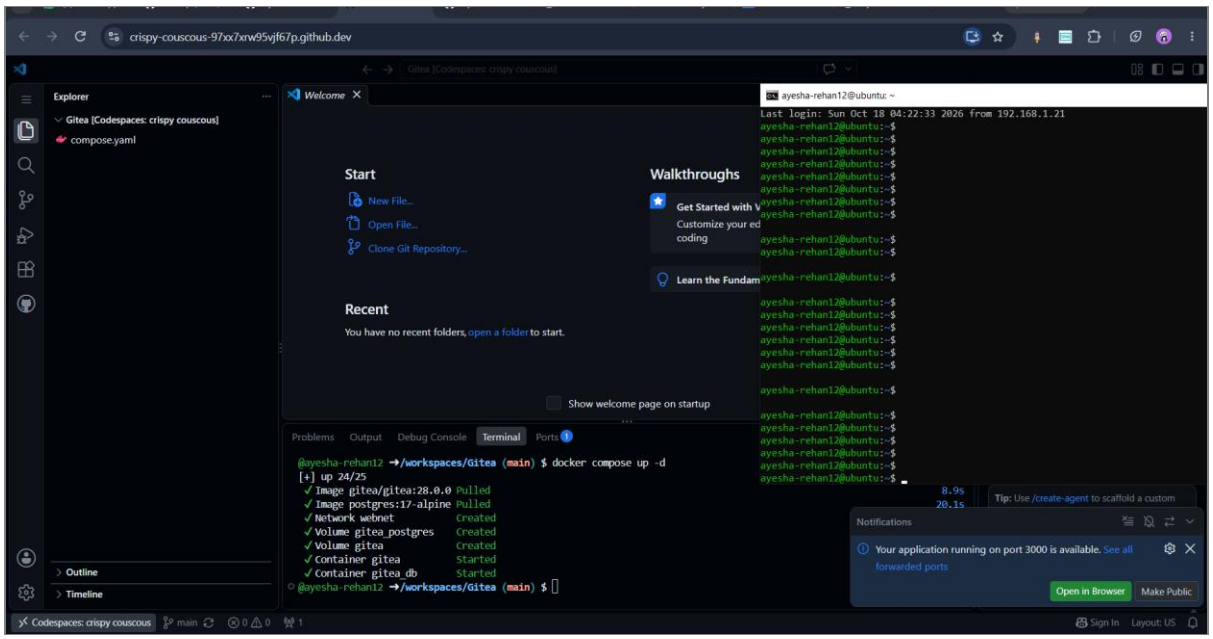


Figure: Containers running after docker compose up -d.

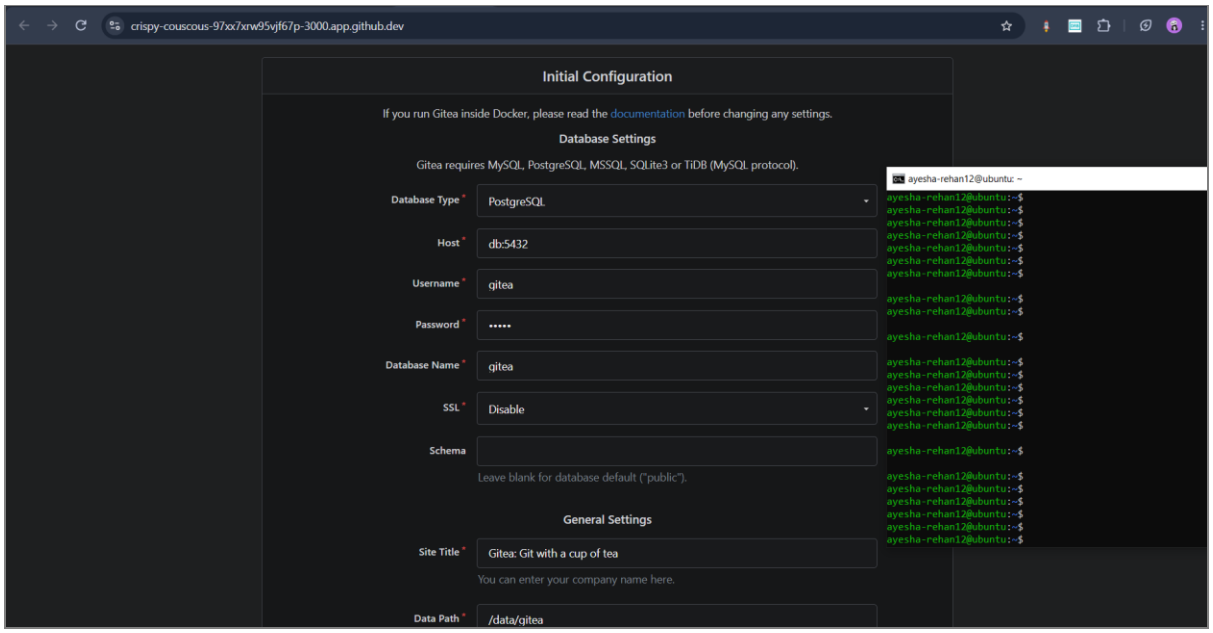


Figure: Gitea install page opened through forwarded port 3000.

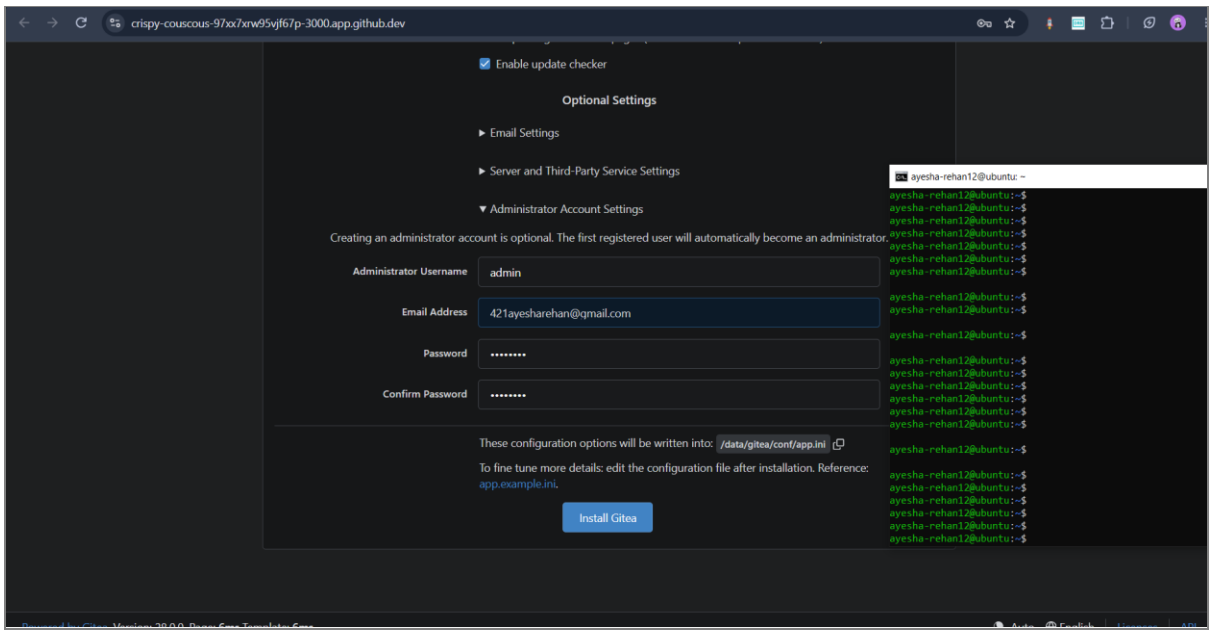


Figure: Gitea installation form with admin account setup.

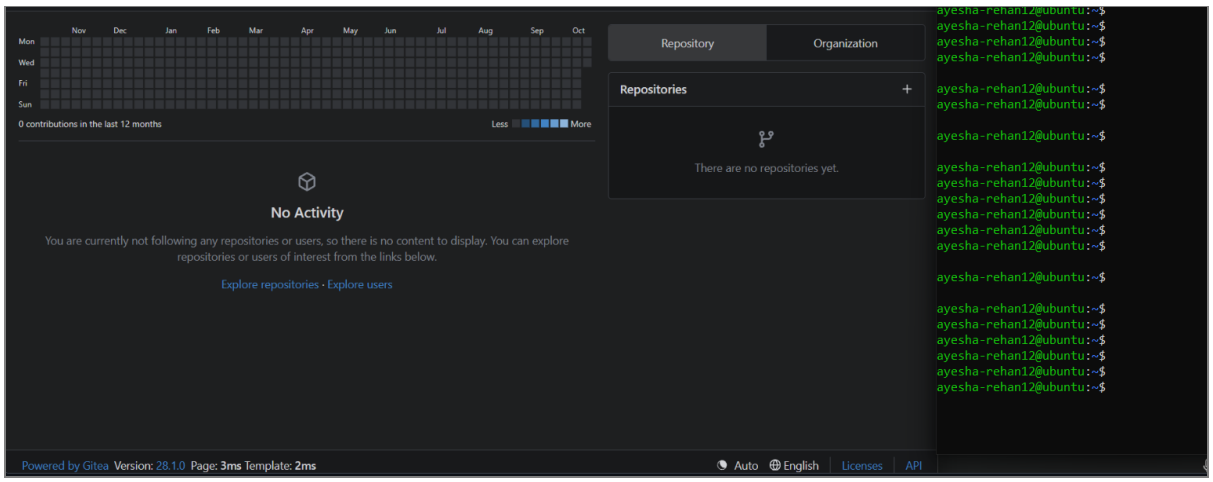


Figure: Gitea dashboard after logging in.

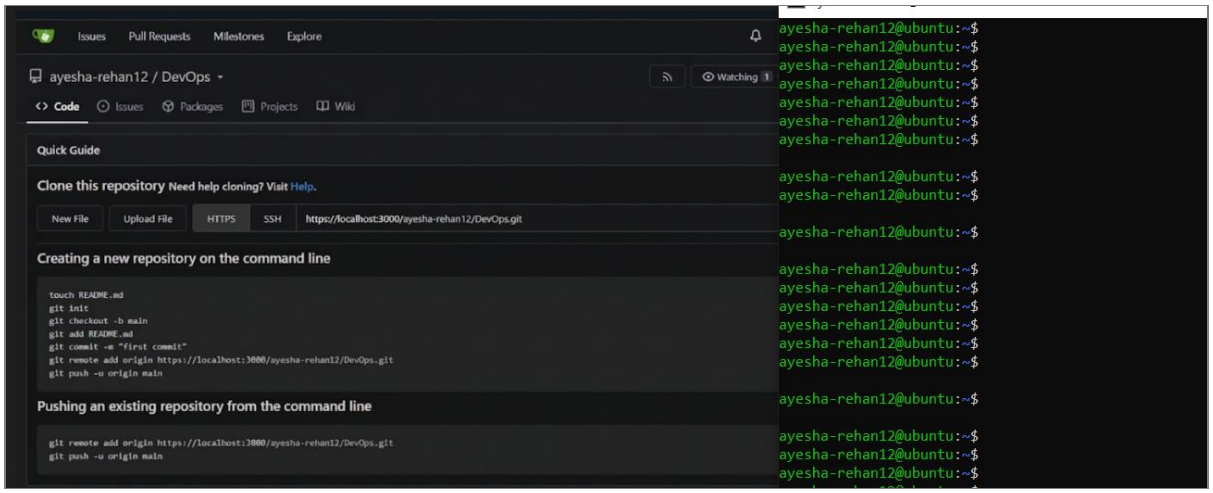


Figure: New repository created in Gitea.

Task 11: Creating a GitHub Pages Portfolio

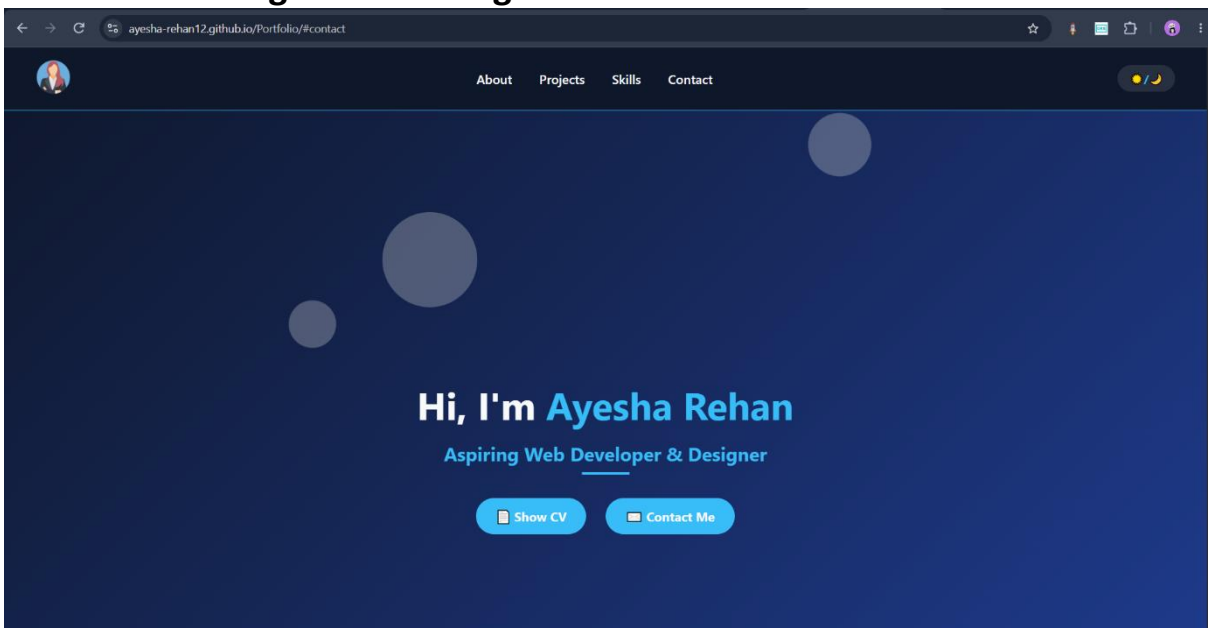


Figure: Live portfolio site opened in the browser.